

DraftRun Pipeline AS IS

Current as of Slice 2.15.6.4.2: Final Gate Attribution Handoff Repair.

This document is the maintained technical map of the current DraftRun generation pipeline. It describes the running system as it exists now, not the target design. Update this file and regenerate the PDF whenever a slice changes DraftRun steps, artifacts, role model usage, context flow, retry/fallback behavior, validation, ranking, revision, or trace semantics.

PDF quick-view copy: docs/architecture/DRAFT_RUN_PIPELINE_AS_IS.pdf.

Regenerate PDF:

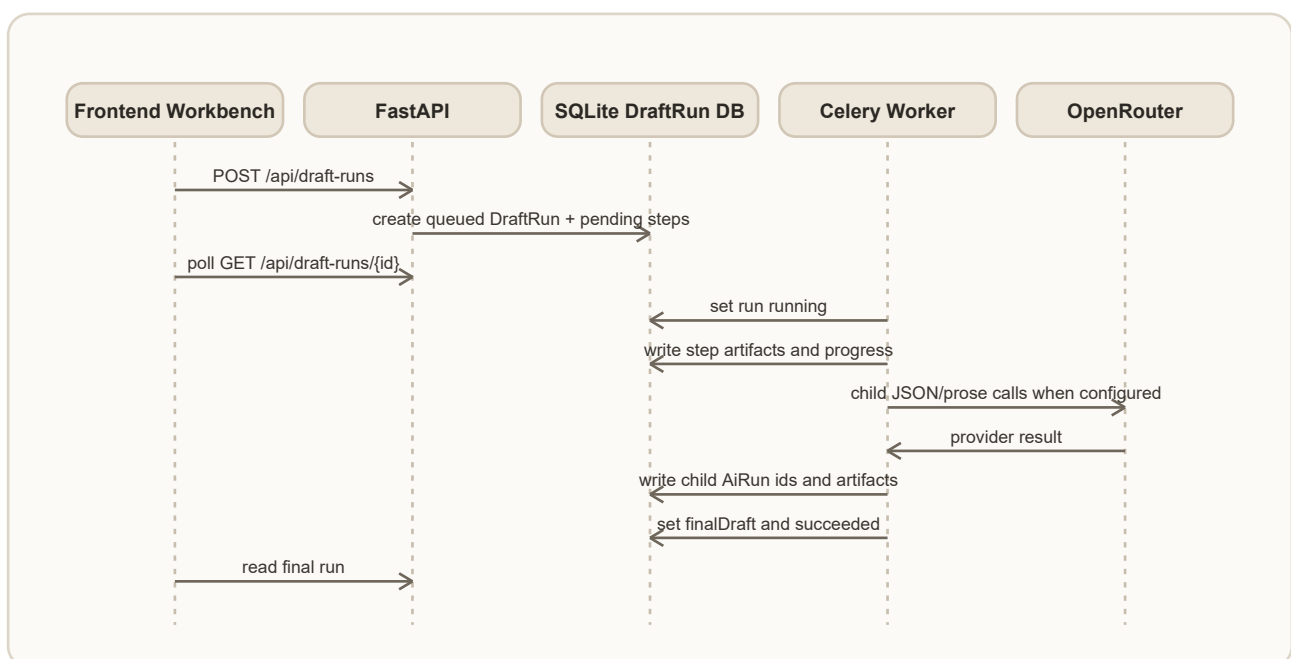
```
python scripts/generate-draft-run-pipeline-pdf.py
```

1. Core concepts

Concept	Meaning	Stored where
DraftRun	Parent orchestration run for one approved post fabula. It owns step order, status, final draft, and child AiRun ids.	var/glavred-draft-runs.sqlite3
DraftRunStep	One logical pipeline stage: context, source intent, public evidence, feasibility, contract, rules, planning, candidates, validation, complete.	draft_run_steps.artifact_payload
AiRun	One model/provider call or audited fallback inside a DraftRun.	var/glavred-ai-runs.sqlite3
SourceLedger	Claim/provenance inventory with allowed use, confidence, warnings, risks, and forbidden inferences.	context and enriched publicEvidence artifacts
EvidenceInterpretation	Editorial interpretation of accepted evidence: implications, tensions, usable examples, limits, forbidden overclaims, reader-value hooks, and rejected uses.	rulePack.evidenceInterpretation, child AiRun trace
PostContract	Locked editorial intent: thesis, audience, CTA, allowed claims, forbidden moves, size contract, topic/fabula obligations.	postContract artifact
RuleRegistrySnapshot	Machine-readable rules and validator bindings derived from contract, ledger, topic, fabula, and publisher rules.	rulePack.ruleRegistrySnapshot
ArticleDossier	DraftRun-local article memory built from artifacts. It is not workspace persistence and not vector storage.	selected step artifacts
ContextPack	Role-specific subset of ArticleDossier passed to LLM calls.	step artifacts and child AiRun.requestPayload
EditorialCritiqueReport	Report-only prosecutor/editor critique of candidate idea strength, tension, author stance, source integration, and reader value.	validation.editorialCritiqueReport, child AiRun trace
AlternativeAngleTournament	Critic-driven challenger route and candidate used to escape a weak local optimum.	validation.alternativeAngleTournament, child AiRun trace
RevisionLoop	Bounded editorial optimization loop: validator repair goals plus editorial goals, old-vs-new dimension scoring, accepted/rejected revisions, and rejected moves.	validation.rankingRevision.revisionLoop

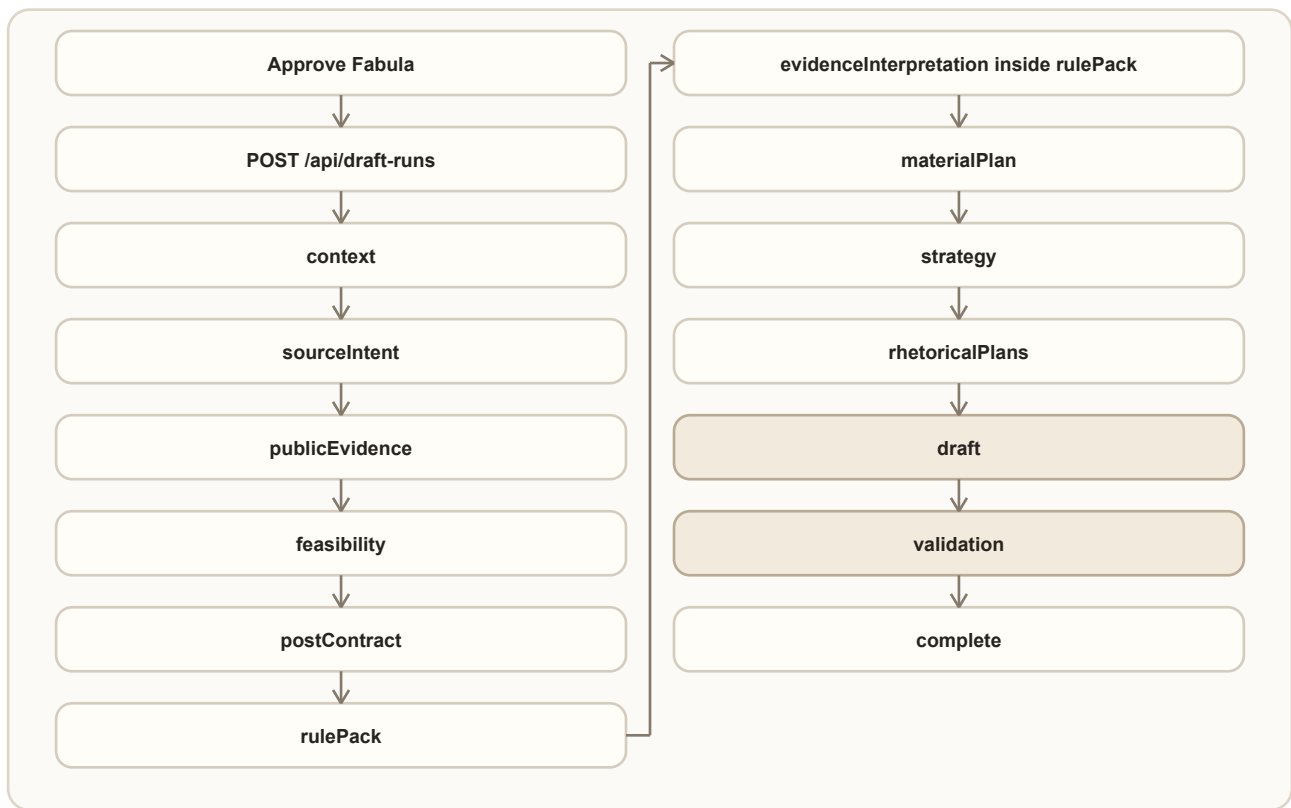
Concept	Meaning	Stored where
FinalQualityGate	Last machine acceptance layer for the delivered final draft as public prose; combines deterministic checks with an independent final-gate model review and may run bounded writer repair cycles if actionable gate findings require it. It separates real missing attribution from diagnostic attribution handoff noise.	validation.rankingRevision.finalQualityGate
DraftRunBudget	Effective research/execution caps derived from Fabula.researchDepth and backend execution mode.	context.draftRunBudget, downstream budget metadata
finalDraft	The selected draft returned to the frontend after validation, ranking, revision loop, and final quality gate.	parent DraftRun

2. Runtime topology



The frontend never treats a live queued/running DraftRun as a reason to call the compatibility `/api/drafts/generate` fallback. Once the parent run exists, the parent run is the source of truth until it succeeds, fails, or is shown as stale.

3. Current step order



The backend enum is:

context -> sourceIntent -> publicEvidence -> feasibility -> postContract -> rulePack ->
 materialPlan -> strategy -> rhetoricalPlans -> draft -> validation -> complete

4. Step-by-step technical flow

The role system is not a separate chat room. Roles interact through persisted DraftRun artifacts. Each LLM call chooses a role model through DraftModelRoleResolver, receives the artifact set and the role-specific ContextPack, writes a child AiRun, and stores its result back into the parent step artifact. The next role reads that artifact instead of relying on hidden conversation state.

Role-aware execution map:

Step	Active role	Model selection	Context passed to the role	Output consumed by
context	none	no model	frontend snapshot only	source intent, ledger, feasibility, DraftRun budget
sourceIntent	research	DRAFT_RESEARCH_MODEL, then default	brief sources, context summary, initial ledger, DraftRun budget	public evidence
publicEvidence search	web search	OPENROUTER_WEB_SEARCH_MODEL	budget-capped research tasks and built search query	evidence synthesis
publicEvidence synthesis	research	DRAFT_RESEARCH_MODEL, then default	accepted public evidence, initial ledger, context	enriched ledger, dossier
feasibility	none	no model	enriched ledger and warnings	contract or blocked completion
postContract	none	no model	brief, ledger, feasibility, size settings	rules, planning, validation

Step	Active role	Model selection	Context passed to the role	Output consumed by
rulePack	none	no model	contract, ledger, publisher/topic/fabula rules	evidence interpretation, validation, revision
evidenceInterpretation inside rulePack	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	enriched ledger, evidence synthesis, public evidence, contract, registry, strategy ContextPack	material plan, dossier, context packs
materialPlan	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	strategy ContextPack, usable evidence candidates, evidence interpretation, contract, registry	draft strategy
strategy	strategy	DRAFT_STRATEGY_MODEL, then default	material plan, rules, contract, strategy ContextPack	rhetorical plans
rhetoricalPlans	strategy	DRAFT_STRATEGY_MODEL, then default, repair, backup	strategy, material plan, claim/rule ids	writer candidates
draft	writer	DRAFT_WRITER_MODEL, then default, repair, backup, domain-safe fallback	writer ContextPack, one rhetorical plan, material evidence, contract, writer generation params	validation and ranking
validation lint	none	no model	candidates, contract, registry, ledger	LLM validation and ranking
validation LLM review	review	DRAFT_REVIEW_MODEL, then default, repair, backup	review ContextPack, candidates, deterministic findings	pairwise ranking
validation editorial critique	critic	DRAFT_CRITIC_MODEL, then default, repair, backup	critic ContextPack, candidates, evidence interpretation, validation findings	trace, dossier, future critique-aware ranking
validation alternative angle	anotherAngle then writer	DRAFT_ANOTHER_ANGLE_MODEL for route, DRAFT_WRITER_MODEL for challenger prose, repair, backup, no fake challenger fallback	initial validation, editorial critique, another-angle ContextPack, writer ContextPack, another-angle/writer generation params	final validation and ranking
validation ranking	review	DRAFT_REVIEW_MODEL, then default, repair, backup	merged candidates, old scorecard, deterministic and LLM findings	revision loop
validation revision	writer	DRAFT_WRITER_MODEL, then default, repair, backup, failed/not-run if no usable JSON	current best candidate, repair goals, anti-regression constraints, writer ContextPack, revision generation params	regression guard and final draft
validation final quality gate	finalGate, then optional writer	DRAFT_FINAL_GATE_MODEL, or independent critic/review/default fallback; DRAFT_WRITER_MODEL for bounded final repair cycles	delivered final candidate, FinalQualityContract, validation reports, critique, evidence interpretation, ledger, contract, rule registry, material plan	accepted public draft or rejected repair trace
complete	none	no model	final DraftRun state	frontend

Practical reading rule: when a step says Role/model handoff: strategy, it does not mean that the strategy model talks directly to the writer model. It means the strategy role writes a structured artifact, then the writer role later receives that artifact plus its own compact context pack.

4.1 context

Purpose: build the normalized local context for the selected work item.

Input:

- approved PostBrief;
- EditorialModel;
- draftContext snapshot from frontend: work item, plan slot, candidate, signal, topic, fabula, publisher rules, author-position evidence, publication-size context.

Processing:

- build_draft_run_context_summary(...) normalizes the frontend snapshot;
- SourceLedgerBuilder builds the initial internal claim ledger.
- DraftRunBudgetResolver combines fabula.researchDepth with DRAFT_RUN_EXECUTION_MODE.

Output:

- contextSummary;
- initial sourceLedger;
- draftRunBudget;
- missingContext and compatibility metadata when links are absent.

Role/model handoff:

- no LLM role is used;
- this step creates the first artifact boundary for later roles;
- downstream research, strategy, writer, and review calls never read the raw frontend workspace directly after this point.

Trace: steps[].stepKey = context.

4.2 sourceIntent

Purpose: convert approved brief sources into an explicit research plan.

Input:

- PostBrief.sources;
- context summary;
- initial SourceLedger.

Processing:

- SourceIntentNormalizer classifies URLs, named sources, human-language requests, proof needs, framing hints, exclusions, and unknown lines;
- ResearchPlanService may call OpenRouter through the research role;
- deterministic fallback preserves original wording when provider planning fails;
- source_research_budgeting caps executable verification tasks and records budgetTrace.budgetSkipped instead of silently dropping them.

Output:

- normalized sourceIntent;
- researchPlan;
- draftRunBudget and budgetTrace;
- child AiRun when OpenRouter is used.

Role/model handoff:

- active role: research;
- model resolver chooses DRAFT_RESEARCH_MODEL or falls back to OPENROUTER_DEFAULT_MODEL;
- the role receives sources, context summary, and the initial ledger;
- it returns a ResearchPlan, not proof;
- publicEvidence consumes the plan and decides what can actually be read or searched.

Trace: sourceIntent step and child AiRun.requestPayload.draftRunStep = sourceIntentResearchPlan.

4.3 publicEvidence

Purpose: execute available public evidence tasks and enrich the ledger.

Input:

- sourceIntent artifact;
- researchPlan;
- current context artifact.

Processing:

- exact URL tasks are read by the URL reader;
- public search tasks call OpenRouter web search only when web tools are enabled;
- retrieval tasks, URL reads, and search result counts are capped by DraftRunBudget;
- disabled or unavailable search tasks remain explicit attempts, not proof;
- relevance guard rejects search-result drift;
- EvidenceSynthesis reconciles accepted public evidence into external claims;
- SourceLedgerExternalEvidenceMerger merges accepted external claims into the ledger;
- accepted evidence items and external ledger claims are trimmed by budget before downstream planning;
- operation progress is persisted during URL/search/synthesis work.

Output:

- PublicEvidenceBatch: attempts, accepted evidence items, warnings;
- EvidenceSynthesis: external claims, decisions, warnings;
- enriched SourceLedger;

- budget metadata: used counts, cap hits, skipped tasks, trimmed evidence/claims;
- initial ArticleDossier and all role ContextPacks.

Role/model handoff:

- public search uses the web-search model/config;
- evidence synthesis uses research.
- web search receives concrete URL/search tasks and built search queries;
- research synthesis receives accepted evidence candidates and decides how they should become external ledger claims;
- downstream roles consume the enriched ledger and dossier, not raw search output.

Trace:

- publicEvidence artifact;
- child AiRun.requestPayload.draftRunStep = externalEvidenceSynthesis;
- child web-search AiRun records when search is enabled.

AS IS limitation:

- public evidence retrieval and synthesis do not write prose directly. Accepted evidence is merged into the ledger here, then interpreted inside rulePack before material planning and writing.

4.4 feasibility

Purpose: decide whether it is safe to generate prose.

Input:

- enriched SourceLedger;
- context warnings;
- evidence availability.

Processing:

- deterministic quality gate classifies the run as feasible, feasible with constraints, needs research, needs human decision, or infeasible.

Output:

- FeasibilityReport;
- if blocked: complete.status = blocked, finalDraft = null, no local fallback.

Role/model handoff:

- no LLM role is used;
- this deterministic gate decides whether later roles are allowed to write prose;
- if blocked, strategy, writer, and review are not called.

Trace: feasibility step.

4.5 postContract

Purpose: lock what the post must preserve.

Input:

- approved brief;
- context summary;
- enriched SourceLedger;
- feasibility result;
- publication-size profile and fabula size intent.

Processing:

- PostContractBuilder resolves title, thesis, audience, value, goal, CTA, platform/date/time, topic/fabula ids, allowed claim ids, qualified claims, forbidden moves, evidence obligations, fabula obligations, risks, and PublicationSizeContract.

Output:

- provider-free PostContract.

Role/model handoff:

- no LLM role is used;
- this step creates the invariant contract that all later roles must preserve;
- strategy may plan within the contract, writer may execute it, and review evaluates candidates against it.

Trace: postContract step.

4.6 rulePack

Purpose: turn the contract and editorial rules into machine-readable rules.

Input:

- enriched context artifact;
- SourceLedger;
- FeasibilityReport;
- PostContract;
- topic/fabula/publisher rules.

Processing:

- rulePack is marked running before provider-backed evidence interpretation, so a long JSON retry path is visible as rulePack, not as stale postContract;
- RuleRegistrySnapshot is compiled first;
- compatibility RulePack is derived from the registry;
- EvidenceInterpretationService converts accepted internal/external evidence into

editorial implications, tensions, examples, limits, forbidden overclaims, reader value hooks, and rejected evidence uses before material planning.

Output:

- ruleRegistrySnapshot;
- compatibility rule-pack fields;
- evidenceInterpretation;
- attempts[] for the interpretation provider/repair/backup path;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- rule compilation itself uses no LLM role;
- evidence interpretation uses the strategy role through DRAFT_STRATEGY_MODEL, then default, repair, optional backup, and finally deterministic fallback;
- this step translates editorial obligations into machine-readable rule ids and then translates evidence into usable editorial meaning;
- later role prompts receive rule ids and validator bindings so they do not work from anonymous prose constraints only;
- material planning and writing receive interpretation artifacts instead of raw citation snippets as their main source context.

Trace:

- rulePack step;
- child AiRun.requestPayload.draftRunStep = evidenceInterpretation when provider interpretation runs;
- artifactPayload.progress.operations[] contains evidenceInterpretation primary/repair/backup attempts while the step is running;
- rulePack.evidenceInterpretation readable section in /ai-runs?runId=...

4.7 materialPlan

Purpose: decide what material is usable before strategy and writing.

Input:

- context summary;
- RulePack and RuleRegistry;
- EvidenceInterpretation;
- enriched SourceLedger;
- PostContract;
- strategy ContextPack.

Processing:

- material planner receives usableEvidenceCandidates;
- the candidate list is capped by DraftRunBudget.maxUsableEvidenceCandidates;
- material planner receives evidenceInterpretation and should prefer implications, usable examples, limits, and forbidden overclaims over raw public snippets;
- it must either choose usable evidence or explain rejection reasons;
- accountability guard rejects empty/unexplained evidence selection;
- retry sequence: primary -> repair -> optional backup model -> emergency deterministic fallback.

Output:

- MaterialPlan;
- availableEvidence;
- rejectedEvidence;
- claimsRequiringAttribution;
- qualifiedClaims;
- attempts[];
- accountability metadata;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL, then default, then repair/backup

where applicable;

- the role receives the strategy ContextPack, usable evidence candidates, ledger claim ids, contract, and registry;
- it must return selected or rejected evidence with reasons;
- strategy and rhetoricalPlans consume the material plan instead of scanning the whole ledger again.

Trace:

- materialPlan step;
- child AiRun.requestPayload.draftRunStep = materialPlan.

4.8 strategy

Purpose: define the main strategy for the post.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- strategy ContextPack.

Processing:

- OpenRouter JSON call or deterministic fallback builds the draft strategy.

Output:

- DraftStrategy: thesis angle, opening move, argument sequence, fabula usage, CTA plan, forbidden moves, tone notes;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL or default;
- the role receives MaterialPlan plus the strategy ContextPack;
- it creates the main route of argument, but cannot change the PostContract;
- rhetoricalPlans expands this strategy into several executable routes.

Trace: strategy step and child AiRun.requestPayload.draftRunStep = strategy.

4.9 rhetoricalPlans

Purpose: build several routes for executing the same contract.

Input:

- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy.

Processing:

- one JSON call returns 2-3 rhetorical plans;
- retry sequence: primary -> repair -> optional backup -> deterministic fallback.

Output:

- RhetoricalPlanSet;
- plan moves, claim ids to use/avoid, required rule ids, CTA route, risks;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: strategy;
- model resolver chooses DRAFT_STRATEGY_MODEL, then repair/backup if JSON shape is invalid;
- the role receives contract, material plan, draft strategy, claim ids, rule ids, and the strategy ContextPack;
- it returns alternative rhetorical plans that the writer must execute;

- the writer must not invent a new route outside these plan artifacts.

Trace:

- rhetoricalPlans step;
- child AiRun.requestPayload.draftRunStep = rhetoricalPlans.

4.10 draft

Purpose: generate candidate drafts from rhetorical plans.

Input:

- request brief;
- context summary;
- RulePack;
- MaterialPlan;
- DraftStrategy;
- RhetoricalPlanSet;
- EvidenceInterpretation;
- writer ContextPack.

Processing:

- one candidate is generated per rhetorical plan;
- the executed candidate count is capped by DraftRunBudget.maxDraftCandidates before provider calls are attempted;
- each provider failure falls back per candidate, not for the whole step;
- candidate publishability guard marks candidates eligible, penalized, or excluded;
- deterministic v1 selection creates an initial scorecard.

Output:

- candidates[];
- selection;
- scorecard;
- child AiRun ids;
- updated ArticleDossier and ContextPacks.

Role/model handoff:

- active role: writer;
- model resolver chooses DRAFT_WRITER_MODEL or default;
- each candidate call receives one rhetorical plan, writer ContextPack, material evidence, evidence interpretation, allowed claim ids, forbidden moves, and size contract;
- each JSON candidate attempt follows the universal retry sequence: primary writer

model, repair prompt, optional backup model, then domain-safe deterministic fallback only if provider attempts are exhausted;

- primary writer attempts use DRAFT_WRITER_TEMPERATURE and DRAFT_WRITER_TOP_P;
- repair and backup attempts use DRAFT_JSON_REPAIR_TEMPERATURE, not the creative writer temperature;
- writer prompts forbid leaking internal artifact names such as SourceLedger, publicEvidence, validators, RuleRegistry, or PostContract as unexplained public dev-jargon;
- candidates write prose, but selection still remains downstream responsibility;
- review receives all candidates plus validation context.

Trace:

- draft step;
- child AiRun.requestPayload.draftRunStep = draftCandidate.

AS IS behavior:

- writer prompts now receive interpreted implications, examples, limits, and forbidden overclaims, so public evidence should shape the argument instead of being pasted as a decorative citation block;
- each provider attempt is visible through child AiRun audit with model role, selected model, attempt label, repair/backup marker, generation params, and safe validation error.

4.11 validation

Purpose: validate candidates, rank them, revise the selected candidate, and choose the final draft.

Input:

- draft artifact;
- context artifact;
- RulePack and RuleRegistry;
- MaterialPlan;
- review/writer context packs through child services.

Processing:

1. deterministic lint checks all candidates;
2. LLM validation checks all candidates and separates actionable findings from observations;
3. editorial critique challenges all candidates for idea strength, tension, author stance, source integration, generic prose, and reader value;
4. alternative-angle tournament may add one critic-driven challenger candidate; both route JSON and challenger-prose JSON use primary/repair/backup attempts, and no fake deterministic challenger is invented when all attempts fail;
5. final validation report is extended when a challenger enters the pool: initial candidate reports are reused, provider-heavy validation runs only for the new challenger, and the reports are merged into one final candidate-pool report;
6. pairwise ranking chooses the best candidate from the merged candidate pool;
7. deep revision loop builds validator and editorial goals, revises, re-runs

deterministic validation, compares previous best vs revised across editorial dimensions, and accepts only targeted improvement without regression. Accepted cycles record concrete reasons such as resolved goals, clear pairwise win, and no deterministic regression;

8. final quality gate builds a `FinalQualityContract` from Fabula research depth, publication kind, post contract, rule registry, and material plan. It checks the delivered candidate with deterministic public-prose rules and an independent `finalGate` model review. If it leaks internal pipeline names, reads like a source dump, loses author stance relative to the contract, or misses reader value, the gate creates bounded final repair instructions for the writer. The number of final repair cycles is controlled by `DRAFT_FINAL_REPAIR_MAX_ITERATIONS` (default 2). A repaired draft is accepted only when deterministic regression and gate findings improve. Attribution warnings are actionable only when they resolve to source-backed claim ids with expected markers that are missing from the final body. Free-text attribution requirements that cannot be resolved become diagnostic handoff noise; if the independent final-gate review passes source integration, that diagnostic noise does not launch final repair;

9. provider-heavy validation sub-operations persist progress as they run; if a late operation fails, the operation is marked failed, safe error details are saved, and the previous best candidate is kept when one exists.

Output:

- deterministic validation report;
- `llmValidationReport`;
- `editorialCritiqueReport`;
- `alternativeAngleTournament`;
- `rankingRevision`;
- `revisionLoop` with editorial goals, dimension scores, rejected moves, and stop

reason;

- `finalQualityGate` with final quality contract, independent review attempts,

final-draft status, public-prose status, source-dump risk, actionable attribution findings, diagnostic attribution noise, attribution review closure, final repair goals, repair cycles, repair result, and gate-level final decision;

- progress with nested validation operations, child `AiRun` ids, failed-operation

status, and safe error details when a provider-heavy sub-operation fails;

- final draft candidate decision;
- updated `ArticleDossier` and `ContextPacks`.

Role/model handoff:

- review for LLM validation and pairwise ranking;
- critic for report-only editorial critique;
- `anotherAngle` for challenger route generation;
- `finalGate` for independent final public-prose acceptance;
- writer for directed revision and final public-prose repair.
- deterministic lint runs first without a model;
- review receives candidates, review `ContextPack`, deterministic findings,

rules, ledger, and contract, then returns report-only findings and pairwise ranking;

- critic receives candidates, critic `ContextPack`, evidence interpretation, and validation context, then returns report-only critique findings and observations;

- anotherAngle receives the initial validation, editorial critique, dossier cards, and another-angle ContextPack, then returns one different contract-safe route;
- writer executes the alternative route into one challenger candidate;
- finalGate receives the delivered final candidate, FinalQualityContract, deterministic gate findings, validation context, evidence interpretation, ledger, rule registry, material plan, and attribution requirement resolution, then returns independent findings, observations, and final repair goals;
- writer may execute bounded final public-prose repair cycles after the revision loop if finalQualityGate returns warning or critical;
- writer receives the current best candidate plus validator repair goals, editorial goals, rejected moves, dossier/context pack, and constraints, then returns revised prose;
- directed revision attempts use DRAFT_REVISION_TEMPERATURE and DRAFT_REVISION_TOP_P; malformed-JSON repair and backup attempts use the JSON repair temperature;
- regression guard checks deterministic health, and review compares old-vs-new by idea strength, tension, reader value, author stance, source integration, structure, and validator health;
- acceptance policy decides whether the revised prose replaces the current best;
- complete receives only the final decision, not the whole model conversation.

Trace:

- validation step;
- child AiRun.requestPayload.draftRunStep = llmValidation;
- child AiRun.requestPayload.draftRunStep = editorialCritique;
- child AiRun.requestPayload.draftRunStep = alternativeAngleRoute;
- child AiRun.requestPayload.draftRunStep = alternativeAngleCandidate;
- child AiRun.requestPayload.draftRunStep = pairwiseRanking;
- child AiRun.requestPayload.draftRunStep = directedRevision.

4.12 complete

Purpose: mark the run result.

Output cases:

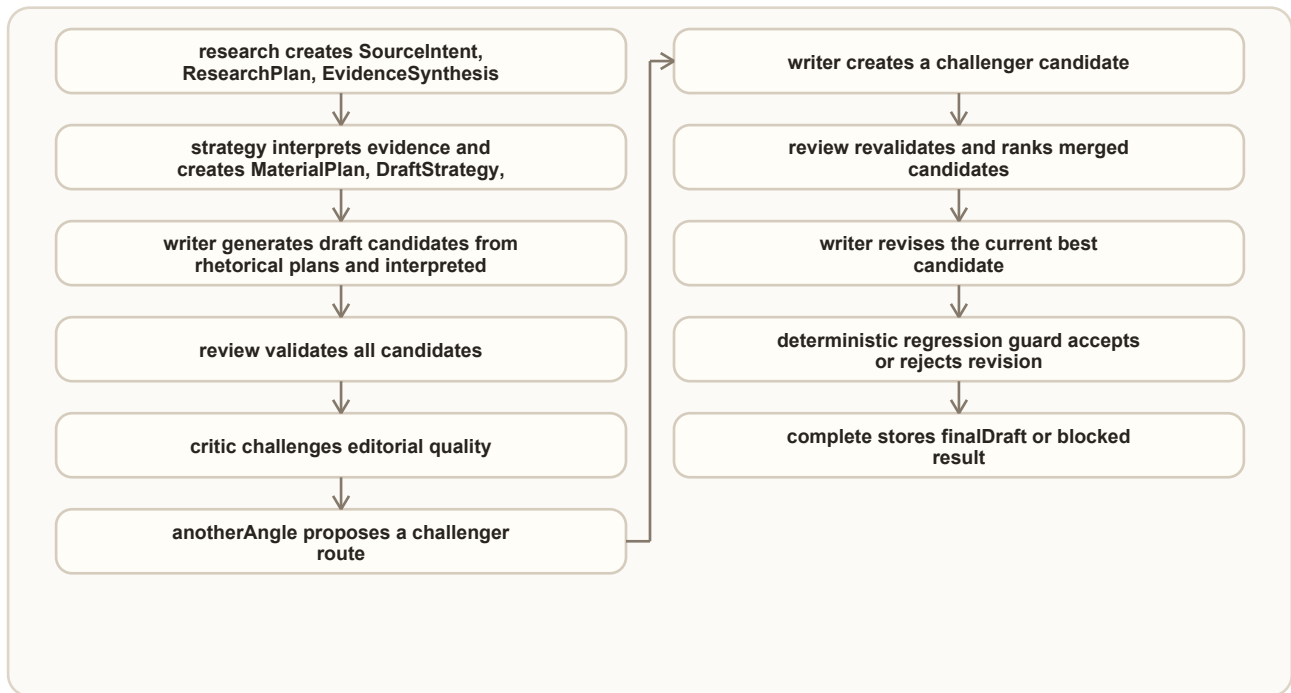
- normal success: parent DraftRun has finalDraft;
- quality-blocked success: parent DraftRun has finalDraft = null and complete.status = blocked;
- failure: parent DraftRun status is failed with a safe error.

Role/model handoff:

- no LLM role is used;
- this step only records the final orchestration outcome for the frontend and trace workbench.

5. Role model usage and handoff AS IS

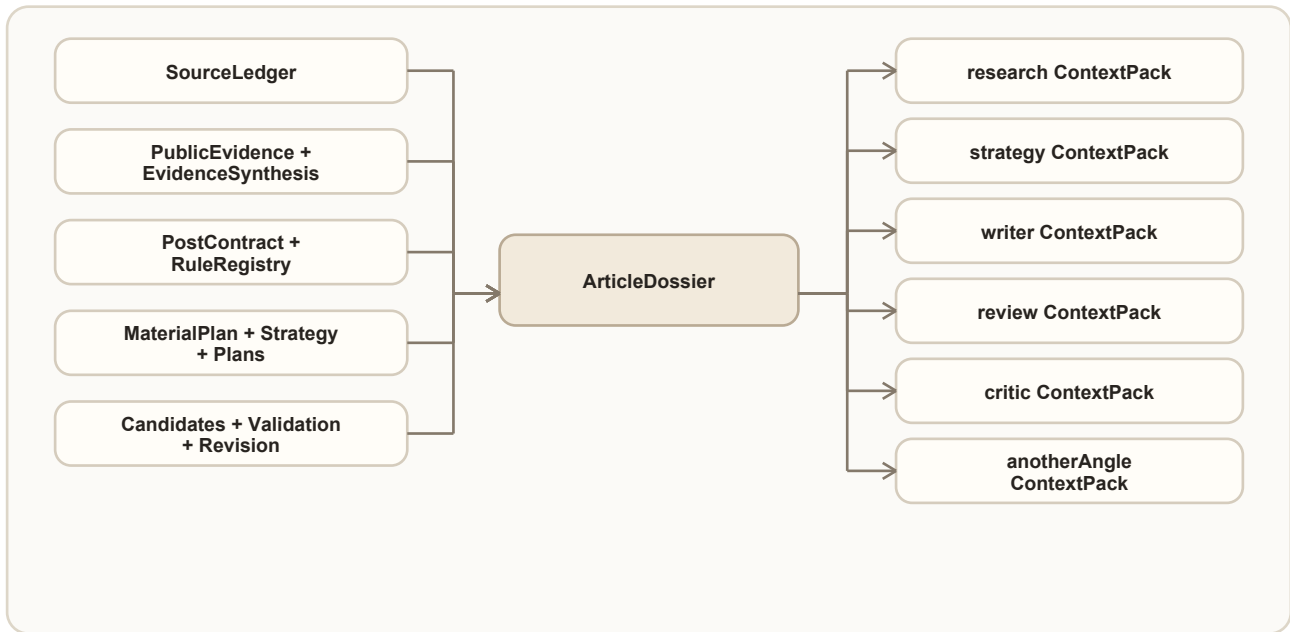
The current role interaction is artifact-mediated. A role does not continue the previous model's hidden conversation. It receives explicit artifacts and a role-specific context pack, then writes a new artifact for the next role.



Role summary:

Role	Used now	Receives	Must not do
research	source research planning and evidence synthesis	source intent, public evidence, context	invent facts or strengthen weak evidence
strategy	evidence interpretation, material plan, draft strategy, rhetorical plans	strategy context pack, rules, contract, material, evidence synthesis	change post thesis or bypass contract
writer	draft candidates and directed revisions	writer context pack, evidence interpretation, material plan, rhetorical plan, repair goals	create new claims or violate forbidden moves
review	LLM validation and pairwise ranking	review context pack, candidates, rules, ledger, validation reports	rewrite prose directly
critic	report-only editorial critique	critic context pack, evidence interpretation, candidates, validation context	rewrite prose or change final selection
anotherAngle	one critic-driven challenger route	another-angle context pack, initial validation, editorial critique, rejected/weak moves	act as technical backup or invent new evidence

6. Context flow



Important AS IS rules:

- raw DraftRun trace is not the normal prompt context;
- ArticleDossier is rebuilt from selected artifacts;
- ContextPack selects up to 12 cards for one role;
- child AiRun.requestPayload stores the role model and context pack it received;
- context packs are local orchestration artifacts, not long-term memory.

7. Artifact catalog

Artifact	Created by	Read by	Debug location
SourceLedger	context/publicEvidence	feasibility, contract, rules, planning, validation	context.sourceLedger, publicEvidence.enrichedSourceLedger
SourceIntent	sourceIntent	publicEvidence	sourceIntent step
ResearchPlan	sourceIntent	publicEvidence	sourceIntent.researchPlan
PublicEvidenceBatch	publicEvidence	evidence synthesis, trace	publicEvidence.items, publicEvidence.attempts
EvidenceSynthesis	publicEvidence	ledger merger, material planning, trace	publicEvidence.evidenceSynthesis
DraftRunBudget	context	sourceIntent, publicEvidence, material plan, draft, validation	context.draftRunBudget, step budget metadata
EvidenceInterpretation	rulePack	material plan, writer, dossier, trace	rulePack.evidenceInterpretation
PostContract	postContract	rules, planning, validation, revision	postContract step
RuleRegistrySnapshot	rulePack	material plan, validation, revision	rulePack.ruleRegistrySnapshot
MaterialPlan	materialPlan	strategy, writer, validation, ranking	materialPlan.materialPlan
DraftStrategy	strategy	rhetorical plans, writer	strategy.draftStrategy
RhetoricalPlanSet	rhetoricalPlans	draft candidates	rhetoricalPlans.rhetoricalPlanSet
DraftCandidate	draft	validation, ranking, revision	draft.candidates[]

Artifact	Created by	Read by	Debug location
ValidationReport	validation	ranking/revision, trace	validation.candidateReports
LlmValidationReport	validation	ranking/revision, trace	validation.llmValidationReport
EditorialCritiqueReport	validation	alternative-angle tournament, trace, dossier	validation.editorialCritiqueReport
AlternativeAngleTournament	validation	final validation/ranking, trace	validation.alternativeAngleTournament
RankingRevision	validation	final draft selection, trace	validation.rankingRevision
FinalQualityGate	validation	final draft selection, trace	validation.rankingRevision.finalQualityGate
ArticleDossier	article memory service	context pack builder, trace	selected step artifacts
ContextPack	article memory service	child LLM services	step artifacts and child AiRun.requestPayload

8. Retry, fallback, and blocked behavior

Area	Behavior
live DraftRun	frontend keeps polling; no compatibility fallback while run is alive
stale DraftRun	diagnostic state after no timestamp progress; not automatic fallback
public search disabled	attempt is notConfigured; it is not proof
malformed JSON planning	repair retry, optional backup model, then deterministic fallback
malformed or empty evidence interpretation	repair retry, optional backup model, then deterministic interpretation fallback
editorial critique provider missing	editorialCritiqueReport.status=not-run; no fake critique
malformed editorial critique JSON	repair retry, optional backup model, then not-run for that candidate
alternative-angle provider missing	alternativeAngleTournament.status=not-run; original candidates continue
malformed alternative-angle route JSON	repair retry, optional backup model, then tournament failed; no deterministic fake angle
alternative-angle candidate provider failure	repair retry, optional backup model, then tournament failed; original candidates continue
validation sub-operation failure	mark nested operation failed, preserve partial artifact, keep previous best if available
material plan ignores evidence	accountability retry, optional backup model, then emergency fallback
unresolved attribution requirement	record diagnostic metadata; not a repair goal unless it resolves to expected source markers that are missing from the final body
candidate provider failure	fallback only for that candidate direction
fallback candidate selection	fallback is diagnostic unless publishability guard allows it
feasibility blocked	run succeeds with finalDraft = null; no local fallback
all invalid candidates	run succeeds as quality-blocked; no local fallback
revision regression	keep previous best candidate

9. How to read a run trace

Open `/ai-runs?runId=<DraftRun ID>` and inspect in this order:

1. context: confirm source signal, candidate, topic, fabula, rules are present.
2. DraftRun budget: confirm research depth, execution mode, effective caps, used counts, cap hits, skipped tasks, and trimmed evidence/claims.

3. `sourceIntent`: confirm research requests were normalized correctly.
4. `publicEvidence`: confirm URL/search attempts, accepted evidence, rejected drift, evidence synthesis, and enriched ledger.
5. `feasibility`: confirm run was allowed or blocked for quality reasons.
6. `postContract`: confirm thesis, CTA, size, allowed claims, and forbidden moves.
7. `rulePack`: confirm rule registry exists and size/evidence rules are present.
8. `evidenceInterpretation`: inside `rulePack`, confirm implications, examples, limits, forbidden overclaims, rejected evidence uses, and attempts.
9. `materialPlan`: confirm selected evidence and explicit rejection reasons.
10. `strategy`: confirm the strategy does not change the contract.
11. `rhetoricalPlans`: confirm the plans are different routes, not new topics.
12. `draft`: compare candidates, source/fallback status, publishability, scorecard.
13. `validation`: inspect deterministic warnings, LLM findings, observations.
14. `editorialCritiqueReport`: inspect idea strength, tension, author stance, source integration, generic-prose risks, and recommended editorial moves.
15. `alternativeAngleTournament`: inspect challenger route, candidate, model attempts, and why it entered or failed to enter final ranking.
16. `rankingRevision`: inspect pairwise winner, revision cycles, accepted/rejected moves.
17. `finalQualityGate`: inspect the final quality contract, independent review attempts/model, final public-prose status, `attributionReview`, `actionableAttributionFindings`, `diagnosticAttributionNoise`, repair goals, accepted or rejected repair cycles, and final draft source.
18. `validation.progress`: inspect nested operations; failed late operations should show safe errors and the final previous-best decision when available.
19. `child AiRun detail`: inspect prompt messages, role model, generation params, context pack, provider result, and sanitized raw response excerpt on JSON failures.

10. Known AS IS limitations

- Editorial critique is active and feeds one alternative-angle challenger plus revision goals, but it is not yet an autonomous multi-agent debate loop.
- `anotherAngle` is active as one challenger route only, not as an autonomous debate loop or general fallback.
- `ArticleDossier` is a local compact memory artifact, not a vector store or cross-run RAG system.
- Failed late provider operations are safely finalized when the Python call returns or raises; a separate infrastructure watchdog is still needed for externally stuck Celery tasks that never return control to the worker.
- The pipeline has detailed traceability, but the main editor UI still receives only one final draft and a compact status.

11. Maintenance rules

- This Markdown document and the PDF must be updated together.
- Regenerate the PDF after every edit:

```
python scripts/generate-draft-run-pipeline-pdf.py
```

- Future DraftRun planning, implementation, diagnostics, evaluation, and autofix work must read this document before changing or judging the pipeline.
- If current behavior differs from this document, update the document first or include the update in the same slice as the behavior change.